



AI for making mistakes

A chess AI that learns from human error, could enable better negotiation skills with humans.
<http://digit.in/mar21-37>



Training counsellors with AI

An AI that behaves like a troubled teen to help counsellors learn to deal with them.
<http://digit.in/mar21-38>

Portable audio troubles

Agent001 has to help a friend with his dull headphones

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his month I ended up helping a friend pick a DAC (Digital-to-Audio Converter). Well he didn't come

straight out and ask for a DAC. Instead he complained about his headphones sounding off when he used them with his smartphone. The same set of headphones when used with his PC would sound just great but the sound would be dull the moment he switched to his phone. His PC has a pretty good motherboard with a dedicated sound circuit, one which has a good AMP and isolated power supply. However, with so many naked circuits around the sound circuit, it's natural for the audio to be a little coloured with all the interference. But here's the thing, not all can tell the difference.

Audio is a rather subjective domain. Grab a bunch of people and make them listen to the same headset and each individual will come up with a separate description of what they've heard. And it's natural since not all ears are the same. And these opinions were all formed of hearing. So there isn't a concrete definition that folks can refer to. An audiophile's dictionary is filled with vague terminologies to describe sounds and to some extent, they do get the job done. Take for example, the term "neutral". It's supposed to

indicate a sound that's devoid of any colouration or modification. Then again, how can you tell if there's no colouration? Where do you draw the line? Sure, something that booms a lot can be termed bass-heavy and sounds made by cymbals are very much what you'd think of when you're talking about treble. The two are nearly at opposite ends of the spectrum for audio instruments, but the spectrum is so wide that one

person's bass-heavy headphones are another man's neutral headphones. Sounds a little silly, doesn't it? But that's the world of subjective audio terminology for you.

So the problem at hand was that the audio signal from the smartphone wasn't as full-bodied as the smartphone. And what we needed was a portable DAC. These things are quite compact and most of them have their own batteries which provide balanced power. Smartphones these days do have DACs of their own which are capable of driving headphones with up to 250 ohms easily. However, the circuitry isn't as full-featured as a portable DAC. There simply isn't enough space within a smartphone to fit a proper circuit. The capacitors alone take up a decent amount of space so instead of fitting a proper audio circuit, you get a larger battery in your smartphone. And the vast majority of people would rather have more battery life than better audio.

A lot of the decent DACs aren't portable. There's the Cambridge Audio DacMagic Plus and Shanling BA1 which have multiple interfaces to connect with not only headphones but also proper desktop speakers and even a home theatre system. But they're not portable. You also have ultraportable devices such as the Fiio A1 and the AudioQuest Dragonfly which are super simple



Fiio Q5S